

POTPRESS · REQUIREMENTS

PotPress requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	PPR-REQ-001	Version	0.5
Status	Draft	Date	2026-09-26
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/potpress		
Source	docs/03-requirements.md @ 77ecec9		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish; R10 target \$990, status met on paper
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R3 at 10 t with a metal stop, R9 pin interlock, R10 \$930, R11 rack 0.8 x 0.8 m and folded rails, R12 lead-free scrap; status from PPR-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Apply Amish's 2026-09-25 decisions (PPR-DDR-001); R10 target \$720, R6 band decided; status from PPR-CAL-001; proposed changes to R3 and R11
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

PotPress requirements

These requirements were checked by calculation at TRL 3 (PPR-CAL-001 v0.3). They are not yet validated with a filter factory and will be revised after co-design sessions (see PPR-PRB-001). None is not met on paper; three are at risk (R2, R7 and R9) and nine are met on paper; see Table 2. Amish's decisions of 2026-09-25 are recorded in PPR-DDR-001 and PPR-DDR-002: the R6 default band, R3 judged at the 10 t working force, the R10 target of \$930 (topped up to \$990 by Amish on 2026-09-26), the R11 rack area of 0.8 x 0.8 m, the pin-presence interlock in R9 and lead-free scrap in R12.

The **reference filter** used throughout is the common flowerpot form: inner rim diameter 280 mm, inner depth 240 mm, wall 15 mm, flat rim about 345 mm across, about 12 L to the brim and about 10 L working volume (dimensions are estimates based on the 280 by 250 mm form reported by [Potters for Peace](#)).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Form the reference filter	Inner rim 280 mm, inner depth 240 mm, wall 15 mm, flat rim; geometry set by parameters in one source file	Model check; later measurement of pressed pots
R2	Even wall	Wall thickness within ± 1 mm around the circumference and from pot to pot; molds self-center to within 0.5 mm	Tolerance stack and guide design; later sectioned pots
R3	Pressing force and frame strength	20 t (196 kN) rated jack; frame, pin, bolts and platen carry 1.5 times the jack rating (294 kN) without yield; deflection between the molds under 1 mm at the 10 t maximum working force, with the molds closing on a metal stop so frame stretch does not set the wall (decided by Amish, 2026-09-25)	Beam and pin calculation; later proof load
R4	Throughput	Press cycle of 6 min or less with two operators, so 50 or more pots per 6 h of pressing	Cycle breakdown; later timed trial
R5	Mold opening	Male mold lifts at least 30 mm clear of the female rim so the mold carriage slides out for loading and demolding; pressing stroke within one jack stroke	Kinematic layout
R6	Flow-rate QC	Rack tests 4 soaked filters at once; T-gauge reads flow over 1 h to 0.1 L/h; repeatability ± 0.1 L/h; acceptance band set per factory (default 1.0 to 2.5 L/h in the first hour, corrected to 25 °C; decided by Amish, 2026-09-25)	Gauge calibration by volume; later repeat tests
R7	Molds made locally	Molds cast from 3D-printed patterns printed on a 250 x 250 mm class printer; finishing with hand tools and a drill press; no lathe over 300 mm swing	Pattern split and finishing review with a foundry
R8	Frame built locally	Standard steel channel and plate; stick welding, drilling and bolting only; heaviest single part 40 kg or less for two-person handling	Part list and mass estimate
R9	Safe operation	Fixed guards on the sides and back, interlocked front gate; jack pump and release outside the guard; a pin-presence interlock prevents pressing unless the load pin is fully home (decided by Amish, 2026-09-25)	Hazard review; later guard check
R10	Affordable	Press plus QC rack \$990 or less in parts (raised from \$600 to \$720 and then to \$930 by Amish, 2026-09-25; topped up to \$990 by Amish, 2026-09-26)	Priced BOM (bom/ bom.csv)
R11	Footprint	Press within 1.0 x 0.7 m floor area and 2.0 m height, with the rail extension folded; QC rack within 0.8 x 0.8 m (relaxed from 0.8 x 0.5 m by Amish, 2026-09-25)	Model check
R12	Product-safe materials	Faces that touch clay or test water are aluminum, food-grade polyethylene or stainless steel; no lead-based paint or oiled release agents on mold faces; molds cast from lead-free scrap (no free-machining alloys; decided by Amish, 2026-09-25)	Material list review

Table 2. Status against each requirement at TRL 3 (from PPR-CAL-001 v0.3, least certain first).

ID	TRL 3 value	Status
R2	Wall ± 1.40 mm as cast; ± 0.43 mm with molds hand-finished to templates; coaxial ± 0.43 mm via the locating lip	At risk
R7	16 pattern segments for a 250 mm printer; cavity finishing without a lathe unproven	At risk
R9	Guards, gate, jack-release interlock and pin-presence interlock in the BOM, not modeled	At risk
R1	Reference filter from PARAMS in <code>cad/src/model.py</code> : 12.30 L to the brim, 9.91 L working	Met on paper
R3	No yield at 294 kN: beams 190 MPa, uprights 54 MPa, 60 mm pin 258 MPa (yield 650), joint bolts 75 MPa. Deflection between the molds 0.61 mm at 10 t (1.84 mm at 294 kN)	Met on paper
R4	5.2 min cycle, 69 pots per 6 h	Met on paper, thin margin
R5	310 mm opening against 270 mm needed; 110 of 150 mm jack stroke	Met on paper
R6	4 stations; 0.1 L is 1.69 mm on the gauge; temperature correction to 25 °C	Met on paper; repeatability not verifiable at TRL 3
R8	Heaviest part 38.9 kg (platen); upright joints bolted, largest frame part 37.4 kg	Met on paper
R11	Press 840 x 655 mm with the rail extension folded (970 mm deployed), 1,806 mm tall; QC rack 780 x 780 mm	Met on paper
R12	Aluminum faces, polyethylene liners, HDPE buckets; lead-free scrap alloy	Met on paper
R10	Priced BOM \$984 (press \$903, QC rack \$81) against \$990	Met on paper, \$6 (0.6 %) margin

Requirement changes decided by Amish, 2026-09-25 (PPR-DDR-002)

- **R3:** deflection judged at the 10 t maximum working force, with the molds closing on a metal stop (was 294 kN; status not met to met on paper).
- **R10:** target raised from \$720 to \$930. The decided bolted joints and hinged rail extension bring the BOM to \$984, so R10 was still not met; Amish topped up the budget to \$990 on 2026-09-26 (see below).
- **R11:** QC rack area relaxed from 0.8 x 0.5 m to 0.8 x 0.8 m; the press footprint is judged with the hinged rail extension folded (status not met to met on paper).
- **R9 and R12:** the pin-presence interlock and lead-free scrap are now part of the requirement text.

Requirement change decided by Amish, 2026-09-26

- **R10:** budget topped up from \$930 to \$990 ("I am ok with the budget top ups"; PPR-DDR-002 v0.2, item 16). The priced BOM of \$984 now meets R10 on paper with a \$6 (0.6 %) margin, which is thin; any price rise moves R10 back to at risk.

■ Assumptions

- Pressing force needed for a well-consolidated wall is not published in the sources found (one low-cost press forms round-bottom filters with a 2 t jack, [Henry, Maley and Mehta, 2013](#)); 5 to 10 t is assumed as the working range and the 20 t jack rating is used as the design load (estimate, to be confirmed with a partner factory).

- Mix follows the RDI-C recipe reported by [Rayner \(2009\)](#); the reference filter needs about 7.5 kg of mix per pot, below the 8 to 9.5 kg factories report (PPR-CAL-001).
- The default flow acceptance band of 1.0 to 2.5 L/h follows Nicaraguan practice and the Potters for Peace range of 1.5 to 2.5 L/h; each factory sets its own band.
- Allowable stress in structural steel is taken as 165 MPa (S275 with a factor of about 1.67).